Importing the barrel/parabolic limit machinery of de Dios Pont into Direction 2 of the Schiffer program

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Why this paper is relevant

In *Convex sets can have interior hot spots* (arXiv:2412.06344), de Dios Pont disproves the hot spots conjecture for convex sets in high dimensions. The construction lives entirely outside the classical 2D arena, but the *mechanism* it uses to pass to a limit is exactly the kind of machinery Direction 2 of the Schiffer program is missing. Three moves stand out and each is portable:

1. **Step 0 (log-concave lift).** The conjecture is rephrased for log-concave measures $\mathbf{1}_\Omega e^{-V}$ with Ω convex and V convex. The original Neumann problem $-\Delta u = \lambda u$ becomes the Langevin eigenvalue problem

$$L_{\Omega,V} u := -\Delta u + \nabla V \cdot \nabla u = \lambda_{\Omega,V}\, u, \qquad \partial_\nu u = 0 \text{ on } \partial\Omega.$$

   The point is that the limit class of "convex objects" is enlarged from convex bodies to convex pairs (Ω, V).

2. **Step 1 (barrel + parabolic collapse).** A convex pair (Ω, V) is lifted to a genuine high-dimensional convex set

$$F_d(\Omega, V) = \big\{(x, w) \in \Omega \times \mathbb{R}^{d+1} : |w| \le \tfrac{1}{2}\big(\sqrt{d} - V(x)/\sqrt{d}\big)\big\}.$$

   The Brunn–Minkowski / Prékopa–Leindler heuristic $\big(\sqrt{d} - V/\sqrt{d}\big)^d / d^{d/2} \to e^{-V}$ tells you why this is the right shape. The crucial PDE phenomenon (Theorem 2.6) is that the eigenfunctions $\phi_{F_d(\Omega,V)}$ are radial in w, so $\phi_{F_d}(x, w) = \psi_d(x, 1 - |w|^2/\rho_d^2)$, and after the nonlinear change of variable $t = 1 - 2|w|^2/d$ the elliptic equation in $\mathbb{R}^{n+d+1}$ degenerates to the linear *parabolic* equation

$$\partial_t h = \tfrac{1}{8}\big(\Delta_x + \lambda_{\Omega,V}\big) h \quad \text{on } \Omega \times [0, 1],$$

   with Neumann lateral data and initial data $h(x, 0) = \phi_{\Omega,V}(x)$. The boundary ∂F_d collapses precisely onto the parabolic boundary $\partial_h \Omega = (\Omega \times \{0\}) \cup (\partial\Omega \times [0, 1])$.

3. **Step 3 (shielding by transport).** A collar $R_m \setminus R$ with a steep potential $|\nabla V_m| \approx m^2$ converts the diffusion $-\Delta + \nabla V \cdot \nabla$ into the transport $\nabla V_m \cdot \nabla$. This propagates boundary data along the flow of ∇V_m and breaks the strong maximum principle.

The first two are useful tools for limit-based attacks on Schiffer (Direction 2), and the third is a cautionary stress-test.

Recall: Direction 2 and the obstruction

Direction 2 assumes a sequence of *convex non-disc* counterexamples $(\Omega_j, u_j, \lambda_j)$ to Schiffer, normalizes area/centroid, and extracts a Hausdorff limit by Blaschke selection. The known difficulty is that all subsequences may converge to a disc. Inside the class of convex bodies in fixed dimension, the only way out is a stronger local rigidity theorem near the disc.

The lesson from de Dios Pont is that one can avoid this trap by *enlarging the limit class*: allow the limit object to be a convex pair $(\Omega_\infty, V_\infty)$, or even a parabolic problem on $\Omega_\infty \times [0,1]$.

Three concrete import strategies

Strategy A: log-concave compactification of Direction 2

Replace "convex domain" by "convex pair (Ω, V)". Define the *weighted Schiffer problem*: find (Ω, V, u, λ) such that

$$L_{\Omega,V} u = \lambda u \text{ in } \Omega, \quad \partial_\nu u = 0 \text{ on } \partial\Omega, \quad u|_{\partial\Omega} = \text{const}.$$

The disc with $V \equiv 0$ is a solution, and so are rescalings.

Now run Direction 2 on a sequence of *weighted* convex counterexamples. Two new kinds of limits are admissible:

- $\Omega_j \to \Omega_\infty \neq$ disc, $V_j \to V_\infty$: a weighted Schiffer counterexample.
- $\Omega_j \to$ disc but V_j records the rescaled obstruction (e.g. $V_j = -\log \mathrm{Jac}_{F_j}$ for a normalized boundary distortion F_j): the limit is a weighted Schiffer pair on the disc.

The Blaschke-only obstruction ("everything collapses to a disc") is replaced by the cleaner question:

> *Are there nontrivial convex potentials V on the disc such that (B, V) is a weighted Schiffer pair?*

This is a sharp algebraic/PDE question on the disc rather than a soft compactness obstruction, and is exactly the kind of question one would ask in Direction 3 (bifurcation from the disc) but now in a two-parameter family (Ω, V).

Strategy B: parabolic Serrin reformulation via the barrel limit

Suppose (Ω, V) is a weighted Schiffer pair with eigenfunction u and constant boundary value c. Lift to $F_d(\Omega, V)$ and ask whether the high-dimensional convex body inherits an approximate Schiffer property for the unweighted Laplacian. Tracking the two pieces of ∂F_d through the limit of Theorem 2.6:

- The *lateral* side $\{x \in \partial\Omega,\ |w| < \rho_d(V)(x)\}$ sends $u_d|_{\mathrm{lat}} = c$ into "$h(x,t) = c$ for $x \in \partial\Omega,\ t \in [0,1]$".
- The *caps* $\{x \in \Omega,\ |w| = \rho_d(V)(x)\}$ send $u_d|_{\mathrm{cap}} = c$ into "$h(x,0) = c$ on Ω".

Combined with $h(x,0) = \phi_{\Omega,V}(x)$, the cap condition forces $\phi_{\Omega,V} \equiv c$, hence $c = 0$ and $u \equiv 0$. So *barrel-lifting cannot turn a weighted Schiffer counterexample at $\lambda_{\Omega,V}$ into an asymptotic unweighted Schiffer counterexample in high dimension.* This is a genuine no-go statement, and it identifies the obstruction: the cap is the issue.

The conclusion is more interesting read backwards. Drop the cap condition and keep only the lateral Schiffer data. The natural limit object is then:

A function h on $\Omega \times [0,1]$ with

$$\partial_t h = \tfrac{1}{8}(\Delta_x + \lambda_{\Omega,V})h, \quad \partial_\nu h = 0 \text{ and } h \equiv c \text{ on } \partial\Omega \times [0,1].$$

This is an *overdetermined parabolic problem* – a Serrin-type problem for a linear heat operator with a known reaction term. There is a developed parabolic Serrin literature (Magnanini–Sakaguchi, Vogel, Ciraolo et al.) and the rigidity statements there should be checked against this exact equation. If parabolic Serrin forces Ω to be a disc, this is genuine progress: weighted Schiffer (lateral overdetermination only) implies $\Omega = B$ provided one can show that the Schiffer overdetermination on Ω propagates to lateral overdetermination of $h_{\Omega,V}$, which holds tautologically when u itself is the initial datum and the lateral Dirichlet/Neumann data are time-independent.

Strategy C: shielding as a stress-test for local rigidity

The shielding step of de Dios Pont (Proposition 2.13) shows that a non-rigid behaviour ($h_{\Omega,V}$ attaining its max in the interior) can be *built* by extending a base rectangle with a steep transport potential. Try the analogous construction with the lateral overdetermined data of Strategy B: does there exist a convex pair (Ω, V) on which the Schiffer overdetermination holds on $\partial\Omega$ but Ω is not a disc?

There are two outcomes, and both are useful:

- If a shielding-style construction *succeeds*, it produces a weighted Schiffer counterexample. This would be a spectacular surprise and immediately kill Direction 2 as stated.
- If a shielding-style construction *provably fails*, the reason will be a structural feature distinguishing the Schiffer overdetermined data (Neumann *and* Dirichlet on the same boundary) from the hot-spots overdetermination (max location). This structural feature is then the local rigidity mechanism Direction 2 actually needs.

The asymmetric initial data $\phi_{R,\varepsilon V} \approx \sin x$ plays a critical role in 2412 because the heat extension can move a high boundary spot inward. For the Schiffer lateral data $h \equiv c$ on the lateral boundary the analogous question is whether the transport flow can leave c invariant on $\partial\Omega \times [0,1]$ while the initial datum is nonconstant. The Neumann lateral condition $\partial_\nu h = 0$ imposes that the transport vector field ∇V_m be tangent to $\partial\Omega$ near the collar; combined with $h \equiv c$ on $\partial\Omega \times [0,1]$, this is a much more restrictive constraint than in the hot-spots setting and is the natural place to look for obstruction.

Concrete first task

The cleanest deliverable, in the spirit of the program, is:

> Show that if (Ω, V) is a convex weighted Schiffer pair, the heat extension $h_{\Omega,V}(x,t)$ defined by Definition 2.7 of 2412.06344 satisfies the lateral overdetermined parabolic system
>
> $$\partial_t h = \tfrac{1}{8}(\Delta_x + \lambda_{\Omega,V})h, \quad \partial_\nu h = 0, \quad h \equiv c \quad \text{on } \partial\Omega \times [0,1],$$
>
> with $h(\cdot,0) = u$. Then ask whether the available parabolic Serrin / parabolic symmetry results force Ω to be a disc.

If they do, weighted Schiffer is parabolic Serrin and Direction 2 reduces to a clean rigidity statement that does not require an L^∞ collapse-to-disc argument. If they do not, the exact gap identifies the analytic feature Schiffer rigidity must control beyond what is captured by parabolic theory, which is the right next question.

A second-priority task is to test Strategy C on the disc with a small radial potential εV_0 and check whether a transport-collar extension can preserve the lateral Schiffer data: the linearisation is explicit (eigenfunctions of the disc are Bessel), and the analogue of Proposition 2.10 of 2412.06344 should produce or rule out a first-order obstruction.